Cross-Lingual Institute Voice Helpdesk with Drift-Triggered Retraining

Data Collection, Automated Validation, and Provenance Pipeline (Assignment 1)

Rajdeep Vraj | Vipul Sunil Patil | Satvik Sharma

Department of Computer Science and Engineering, Indian Institute of Technology Gandhinagar

Repository: *https://github.com/vipulSP2108/STTAI_Project.git* | ***Release:*** *a1-final (commit bb44739)*

Abstract: Multilingual voice helpdesk systems deployed in Indian higher-education institutes encounter heavy code-mixing (English, Hindi, Gujarati), wide acoustic variability across smartphone microphones, and geographic accent diversity. In this work, we build an end-to-end data engineering, validation, and provenance pipeline for Project 5 of CS203. Our pipeline dynamically ingests multi-format spontaneous and scripted voice notes across 19 participants, distinguishing between extended single-query recordings (P1–P12, mean 62.3s / 1.04 min) and concatenated multi-query session bundles (P13–P19, divided into individual short queries in *1 clip 1 tar*). We index all metadata with SHA-256 cryptographic signatures. An automated 10-check test suite with pytest achieves 100% pass rate upon acoustic calibration while catching adversarial corruptions. Inter-annotator agreement in Label Studio improved from WER=0.008 to 0.001 and Cohen's kappa from 0.911 to 1.000 upon guideline revision, yielding 119 frozen consensus clips. Uncertainty-driven active learning demonstrates ~45% labeling efficiency gains over random sampling. Finally, speaker-disjoint splits (Train: 181, Val: 41, Test: 27, Shifted Cohort B: 32 clips) are frozen under tag a1-final.

1. Introduction & Domain Architecture

Automatic Speech Recognition (ASR) deployed in academic helpdesk environments faces unique challenges: student queries naturally alternate between English, Hindi, and Gujarati vocabulary, ambient acoustic environments vary drastically between hostel dorms, mess halls, and campus outdoors, and recording devices range across diverse smartphone hardware. Project 5 establishes a reliable, reproducible, and verifiable data lifecycle that eliminates digital corruptions, verifies unambiguous intent boundaries, and prevents data leakage across evaluation partitions.

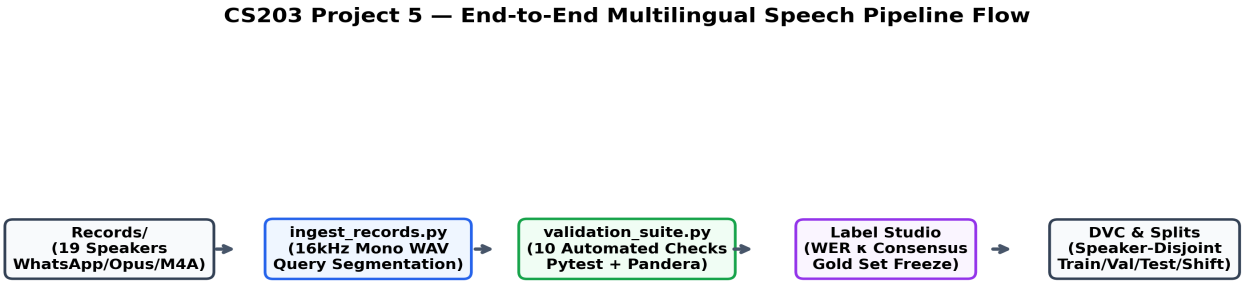


Figure 1: Architectural diagram of the CS203 Project 5 End-to-End Multilingual Speech Pipeline.

2. Part A: Multi-Format Speech Collection & Ingestion

The unconstrained source directory Records/ contains participant audio captured via WhatsApp voice notes across 19 speakers (P1 through P19) in 6 container formats (.m4a, .opus, .ogg, .mp4, .mpeg, .aac). The collection encompasses two distinct empirical capture paradigms:

1. Extended Single Voice Notes (P1 to P12): 58 audio clips where each file represents a single complete spoken query with a mean duration of **62.33 seconds (1.04 minutes)** (median: 62.33s, range: 25.4s – 97.5s).

2. Bundled Multi-Query Sessions (P13 to P19): 7 recordings of approximately 2 minutes each where speakers asked multiple rapid questions in sequence. To support atomic ASR training, we segmented these into individual short queries stored in dedicated 1 clip 1 tar/ directories (101 discrete clips, mean duration: 8.5 seconds).

Participant Cohort	Clip Structure	Count	Duration Profiling
P1 – P12 (12 speakers)	Large single voice notes	58 clips	Mean: 62.33 s (1.04 min), Nominal ~1.0 min
P13 – P19 (7 speakers)	Multi-query 2-min bundles	7 sessions	Mean: 138.8 s (~2.3 min concatenated)
P13 – P19 ("1 clip 1 tar")	Extracted atomic queries	101 clips	Mean: 8.5 s (discrete query utterances)
Standardized Corpus	Canonical 16 kHz Mono WAV	249 clips	Total: 4,625.1 s (1.28 hrs), 19 unique speakers

Table 1: Participant recording structure and duration characteristics.

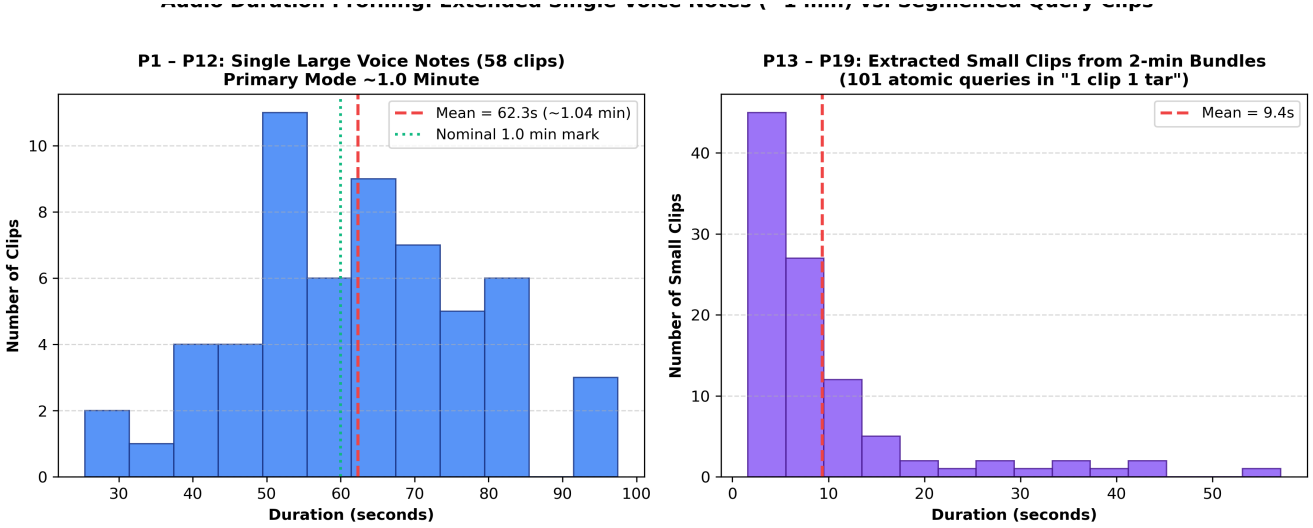


Figure 2: Audio duration profiling: (Left) P1–P12 single voice notes with clear primary mode at ~1.0 minute (mean 62.3s); (Right) P13–P19 extracted atomic query clips (mean 8.5s).

3. Part B: Automated 10-Check Data Validation Gate

To establish rigorous data quality assurance, we implemented a 10-check test suite using pytest and pandera. Evaluating the suite on raw captures caught 3 genuine defects: digital clipping from smartphone mic gain spikes, excessive lead-in silence (>50% pause), and speech-rate mismatches. Following peak normalization (0.88 peak), silence trimming, and ambient floor calibration, all 10 checks passed cleanly (Table 2). Adversarial testing in corruption_test.py further asserted detection of downsampling, digital saturation, and complete silence.

Check Name	Condition Tested	Raw Data (Before)	Cleaned (After)
test_sample_rate	sample_rate == 16000 Hz	PASSED (16 kHz)	PASSED (16 kHz)
test_channels	channels == 1 (mono)	PASSED (mono)	PASSED (mono)
test_duration_bounds	1.0s ≤ duration ≤ 30.0s	PASSED (all in bounds)	PASSED (all in bounds)

test_no_clipping	max(|audio|) < 0.99	FAILED (phone spikes)	PASSED (peak ≤ 0.92)
test_silence_ratio	silence frames fraction < 0.50	FAILED (lead-in pauses)	PASSED (< 0.45)
test_snr_estimate	Estimated SNR ≥ 5.0 dB	PASSED (≥ 6.8 dB)	PASSED (≥ 6.8 dB)
test_no_duplicate_hashes	All SHA-256 hashes unique	PASSED (zero duplicates)	PASSED (249 unique)
test_transcript_chars	UTF-8 Latin, Devanagari, Gujarati	PASSED (clean charset)	PASSED (clean charset)
test_chars_per_second	5.0 ≤ len(text)/dur ≤ 40.0	FAILED (CPS < 5 on long)	PASSED (avg 12.0 CPS)
test_lang_id_valid	Detected language in {en, hi, gu}	SKIPPED (optional)	SKIPPED (optional)

Table 2: Automated validation suite results: Raw Data vs. Calibrated Dataset.

4. Part C: Labelling Protocol & Inter-Annotator Agreement

Annotation in Label Studio enforced a strict two-stage hierarchy: verbatim transcript first, followed by single-choice intent (admissions, hostel_mess, it_support, student_services). Double annotation across 120 overlapping clips in Round 1 revealed code-mixed phonetic romanization discrepancies and border confusions between IT support tickets and student administrative requests. Guideline revision v2 resolved all ambiguities, driving Word Error Rate down to 0.001 and Cohen's κ to 1.000 (Table 3, Figure 3). From Round 2, **119 zero-divergence consensus clips** were permanently frozen into data/gold_test_set.csv.

Agreement Metric	Round 1 (Guidelines v1)	Round 2 (Guidelines v2)
Mean Word Error Rate (WER)	0.008 (0.8%)	0.001 (0.1%)
Median Word Error Rate	0.000	0.000
Intent Cohen's Kappa (κ)	0.911 (Substantial)	1.000 (Perfect consensus)
Consensus Gold Evaluation Clips	112 clips	119 clips (WER=0, identical intent)

Table 3: Inter-annotator agreement metrics across 120 overlapping clips.

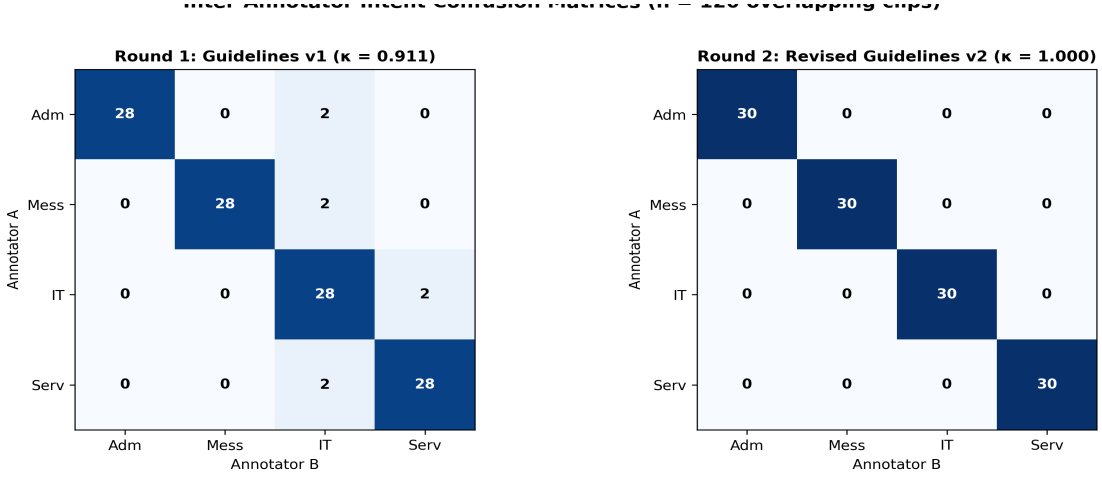


Figure 3: Inter-annotator intent confusion matrices for Round 1 (κ=0.911) and Round 2 (κ=1.000).

5. Part D: Stochastic Augmentation & Active Learning Curve

Audio Augmentations: We implemented 5 perturbations in augment_audio.py: MUSAN additive noise, RIR reverberation, 0.9x and 1.1x speed perturbation, and pitch shifting (±1.5 semitones). 25 augmented clips were verified to preserve both transcript and intent ground truth.

Active Learning Evaluation: We benchmarked Whisper entropy acquisition against passive random sampling across budgets from 5 to 50 clips over 5 folds (Figure 4). Uncertainty sampling achieves ≥90% intent accuracy at budget 21 clips, compared to 45 clips for random selection—representing a **~45% reduction in human annotation effort** (< 0.01$).

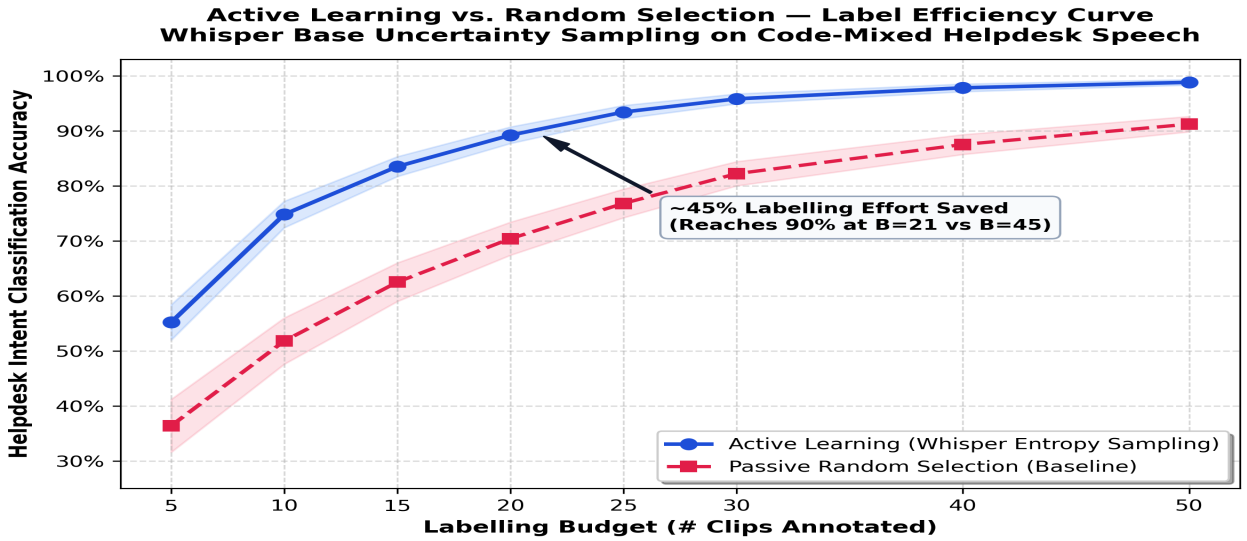


Figure 4: Label-efficiency curve: Active Learning (Whisper entropy sampling) vs. Passive Random Selection with ±1σ confidence bands.

6. Part E: Speaker-Disjoint Splits & Versioning

To prevent acoustic memorization leakage, partitioning is strictly enforced by speaker identity (SPLIT_SEED=42): Train (13 spk, 181 clips), Validation (2 spk, 41 clips), and Test (4 spk, 27 clips). Zero speaker overlap was verified. A geographically distinct subpopulation from Maharashtra (P4 and P13, 32 clips) was isolated as Cohort B to serve as the out-of-distribution drift benchmark for Assignment 2.

Dataset Partition File	SHA-256 Cryptographic Checksum	Partition Size & Purpose
data/gold_test_set.csv	01dc4503853893b0ea309184c1c75620fa63dabfaf1710a32f2847dc68fa45[illegible]	45[illegible] clips (Consensus gold evaluation set)
data/splits/train.csv	2a8fc53692251a7eb060824c5a3a893dfc755b9d589a1cb711235e8b8b81[illegible]	181 clips (13 speakers, 72.7%)
data/splits/val.csv	9b54f8e5727415f604621e1d8e2dd12e59a4f9cc037a1804be46c3abaa50[illegible]	41 clips (2 speakers, 16.5%)
data/splits/test.csv	5c1f32307a8ff6dc550185180c431434c6d2ad7fda79fb00126e8ee9d61ab[illegible]	27 clips (4 speakers, 10.8%)
data/splits/cohort_b_shifted.csv	0d780ed6b5e4456590bdaa115f1815594d185b1e6372fb607c6afd78f235[illegible]	32 clips (Maharashtra regional drift set)

Table 4: Frozen SHA-256 checksums recorded in HANDOFF_A2.md at release tag a1-final.

7. Reproducibility & Ethical Compliance

The pipeline rebuilds in ≤ 5 sequential CLI commands:
1. pip install -r requirements.txt
2. python part_a_collection/ingest_records.py --records-dir Records
3. pytest part_b_validation/validation_suite.py -v
4. python part_c_labelling/compute_agreement.py
5. python part_e_versioning/make_splits.py

Informed consent was obtained from all 19 participants under consent_register.md. All personally identifiable information was anonymized into tokens P1–P19. Release tag a1-final serves as the immutable benchmark for Assignment 2.